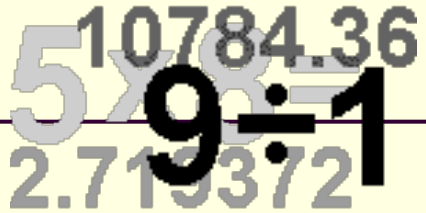
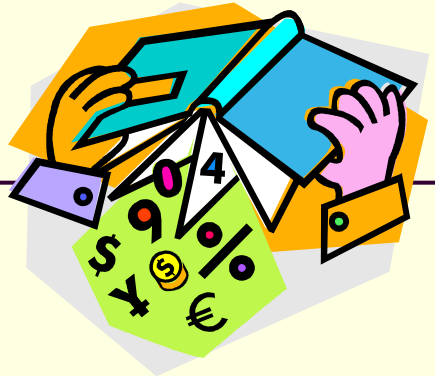
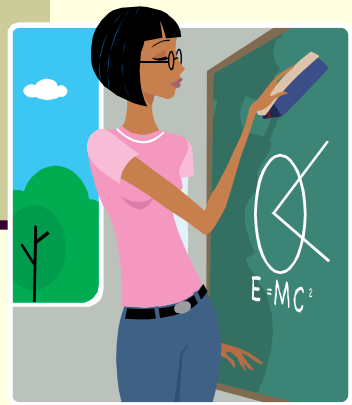




Teaching Approaches and Learning Theories (Part 2)



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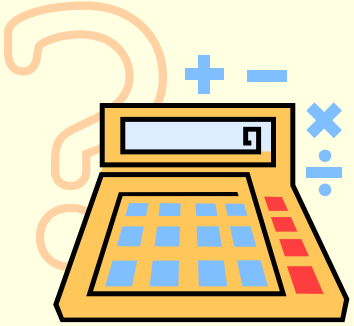


Q1.

Today's Lecture

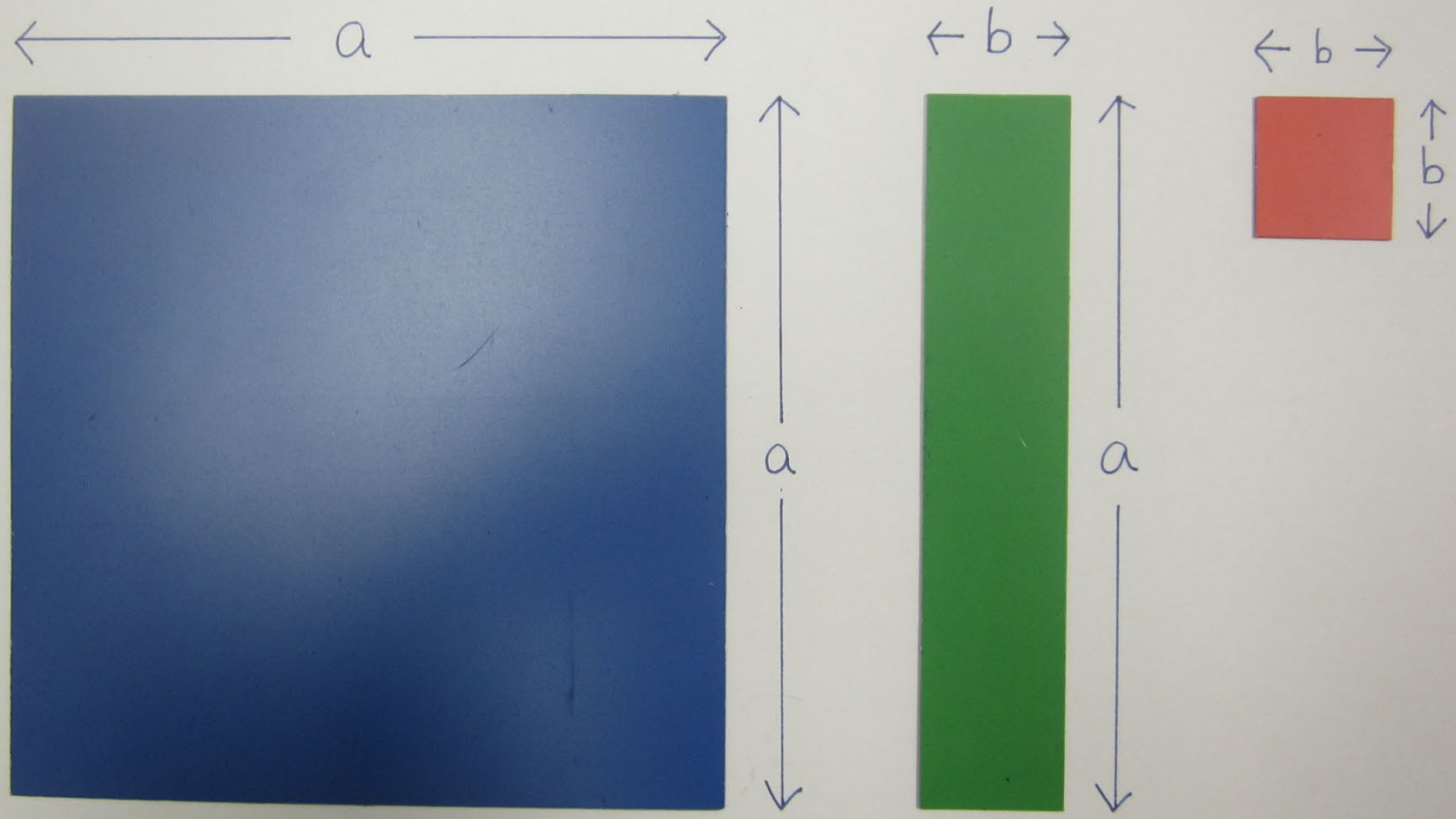
- One more method to develop math concepts
- How to develop procedural skills and positive attitudes (or dispositions)

Key Lesson 1

- _____ **is usually more effective** than proofs to convince students

How to teach $(a + b)^2 = a^2 + 2ab + b^2$

- _____ **C-P-A (concrete-pictorial-abstract) approach** (hyphens signify links)
- Start with **concrete** manipulative, e.g. algebra tiles **(let students play with the tiles)**
- Dimensions of big square tile are a by a, so area of big square tile = a^2
- Dimensions of small square tile are b by b
- Dimensions of rectangular tile are a by b
- Draw the tiles in the slide after next **(pictorial)**


a
a
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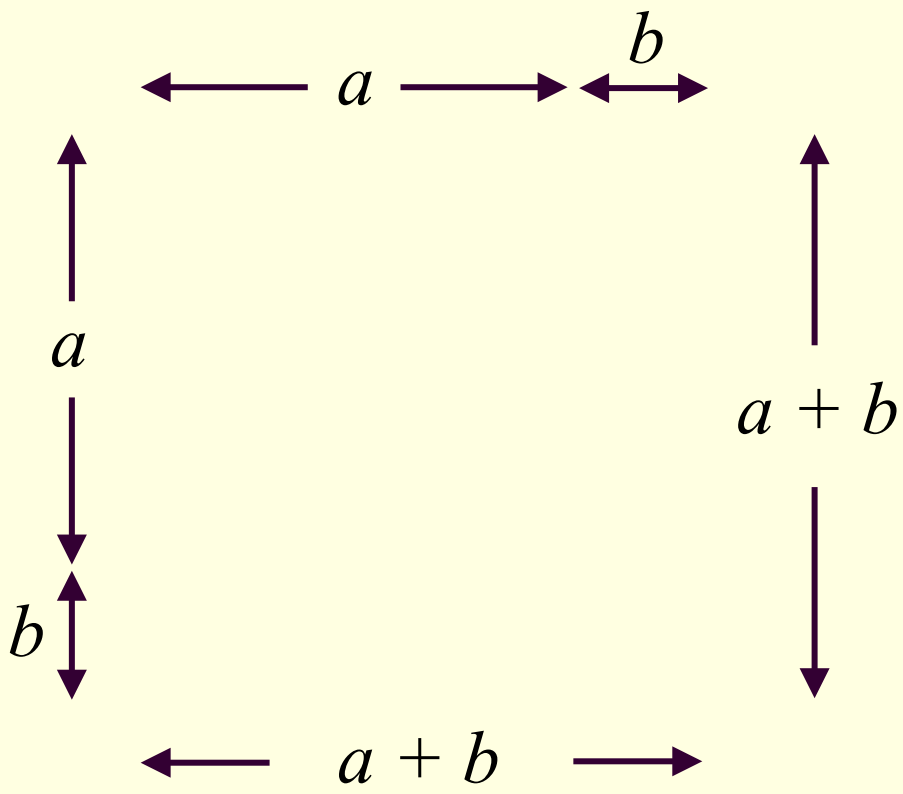
How to teach $(a + b)^2 = a^2 + 2ab + b^2$



- So where does _____ come from?
- Students realise $a^2 + b^2$ not enough
- **Visual representation** can **explain** better than abstract procedure

Bruner's C-P-A Approach

- _____ approach: go from **concrete** algebra tiles, to **pictorial** representation of tiles, to **abstract** identity $(a + b)^2 = a^2 + 2ab + b^2$
- C-P-A is **not using 3 different methods** (hyphens indicate the links between stages)
- Concept development or teaching for understanding is **not to start with proving**, because proofs are abstract
- Teaching for understanding can be **suitable for** _____ **readiness students** because we are not starting with proofs

Extended C-P-A Approach

- At higher level, may not find suitable concrete manipulative or pictorial representation
- Recall guided-discovery worksheet on primes starts with **concrete examples of factors of 1 to 20**, before abstracting out concept and definition of primes
- _____ **C-P-A Approach:** Concrete can mean concrete manipulative or **concrete examples** using numbers; may or **may not have pictorial representation**

Extended C-P-A Approach

- Primary schools: two real apples (concrete objects) → pictures of two apples (pictorial) → **numeral ___** and word 'two' (**abstract**)
- Secondary schools: concrete can mean **concrete examples using numbers, e.g. ___** → pictures (if any) → algebra (abstract)
- University: **algebra (has become concrete!)** → what is abstract is abstract algebra (e.g. rings)
- What is **abstract** at one level will usually become **concrete** at a higher level

Extended C-P-A Approach

- At secondary school level, concrete examples usually involve **using** _____
- Another example: how to teach the product law of logarithms, i.e. $\log_a x + \log_a y = \log_a xy$?
- Do you start with stating law and proving it?
- **Proofs are _____ (last stage of C-P-A)**
- How to use Extended C-P-A approach to teach?

Key Lesson 2

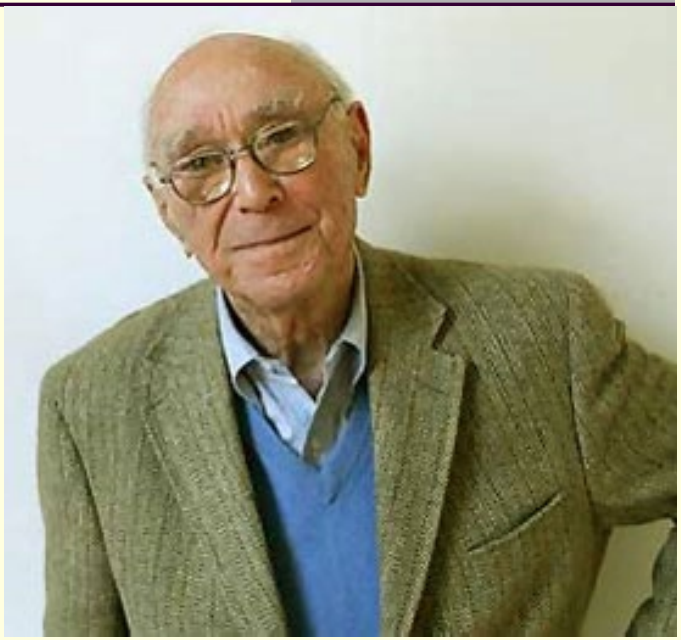
- Extended C-P-A Approach[1]: **Always go from** _____ (e.g. concrete manipulative or concrete examples) **to abstract**

[1] Yeo, J. B. W., Choy, B. H., Lim, L. G. P., & Wong, L. F. (2019). Innovative pedagogical practices. In T. L. Toh, B. Kaur, & E. G. Tay (Eds.), *Mathematics education in Singapore* (pp. 165-193). Singapore: Springer.

How to teach concepts (summary)

- Some teaching approaches include:
  - Bruner's guided-discovery approach
  - Bruner's C-P-A approach
  - Teacher explanation with questioning (more in tutorials)
- These approaches ____ **mutually exclusive**, e.g. Bruner's guided-discovery approach usually goes from concrete to abstract, and teacher explanation may also include guided-discovery

Jerome Bruner



- American psychologist
- 1 Oct 1915 – 5 Jun 2016
- How old did he live? ___
- Piaget (84), Pavlov (86), Skinner (86), Gagné (85), Thorndike (74)
- Search in Wikipedia to find out more about these psychologists and their theories

Pentagon Model: Teaching Skills

- Traditional rule-based teaching based on **theory of** _____ by Thorndike, Skinner, Gagné
- Stimulus will result in a response: establish correct **stimulus-response** (S-R) bonds
- Pavlov noticed that dogs would salivate when food were delivered to them; so he rang a bell before delivering food to his dog; soon his dog associated bell ring with the food, and it would salivate when bell rang without seeing the food

Pentagon Model: Teaching Skills

- _____: stimulus (bell ring) produces response (salivate)
- But human brains more complicated than dog brains, although conditioning still works on humans to some extent
- Moreover, **theory of constructivism** suggests that there are other forces at work: bonds / links in human brain are not simply stimulus-response (S-R) bonds, but **meaningfulness** plays a part

Pentagon Model: Teaching Skills

- Nevertheless, theory of behaviourism still provides a good method for teaching skills, although there is a need to **modify** certain aspects based on theory of constructivism
- Many people think that teaching skills is easy: just show students some worked examples and let them practise
- Few people actually realise you need to plan _____ of worked examples, practice questions and homework questions

Pentagon Model: Teaching Skills

- **Theory of behaviourism**: students **master one skill** _____, before learning new ones
- "I do. We do. You do."
- Teacher shows worked example on whiteboard; teacher does _____ worked example together with students on whiteboard (i.e. asks students questions to guide them); students do a **similar** practice question

Pentagon Model: Teaching Skills

- **Learning theories are _____ for you**: if ST has understood theory of behavourism, she would not have wasted two whole periods
- But there are two problems with "I do. We do. You do."
- First, **theory of constructivism** tells us that we should not just talk to ourselves: **"I do" is not so effective** to help most students construct their own knowledge meaningfully

Pentagon Model: Teaching Skills

- Secondly, you don't have so much teaching time
- In practice: **"We do one. You do _____."**
- Teacher does Worked Example 1 (WE1) with students on whiteboard (i.e. asks students qns)
- **Give students a few practice questions** to do: first question should be _____ to WE1 (how similar is similar?)
- Depend on students' readiness: for low readiness students, **just change numbers** in WE1

Pentagon Model: Teaching Skills

- **Leave solution for Worked Example 1 on whiteboard** so that students can follow working for Practice Q1 (don't be surprised some students can't even follow working)
- Practice Q2 onwards should have _____ **in structure or difficulty level**
- If students have difficulty with Q2 onwards, **guide** individual students, or **guide** whole class verbally, or on whiteboard if necessary

Pentagon Model: Teaching Skills

- How many practice questions for each WE?
- It all depends on the type of questions
- Basically, _____ **WE2** when you reach 'practice question' that most students **don't know how to do**, or how to present solution
- Then repeat cycle of one worked example with a few practice questions
- Possible to have only one practice question for a worked example because no other variety

Pentagon Model: Teaching Skills

- Should you make students copy solutions of worked examples?
- What is the _____ of copying solutions?
  - Internalise while copying?
  - Will students do that? Or will they just copy blindly? Esp. when you don't have so much time
- In fact, copying solutions of worked examples is **very time consuming**; end up **not enough time for practice**

Pentagon Model: Teaching Skills

- But some teachers want their students to have **'model solutions'** to study
  - If worked examples are from textbooks, 'model solutions' already in textbooks: why copy again?
- This is what you can do if worked examples not from textbooks :
  - First practice question similar to worked example: so **student's solution for first practice question** will be '**_____ solution'** to study

Pentagon Model: Teaching Skills

- Should homework questions be different from worked examples and practice questions?
- Some STs think that homework questions should be the more difficult or challenging qns
- Then most students will not do your homework because they can't even start
- **One main purpose of homework** is to for students to _____ **what they have learnt in class**

Pentagon Model: Teaching Skills

- Hence, homework questions should be **similar in** _____ to worked examples and practice questions
- But you can add one or two unfamiliar or more challenging questions **at the end** (if you put this as Q1 and the students don't know how to do, many of them will not continue): another purpose of homework

Key Lesson 3

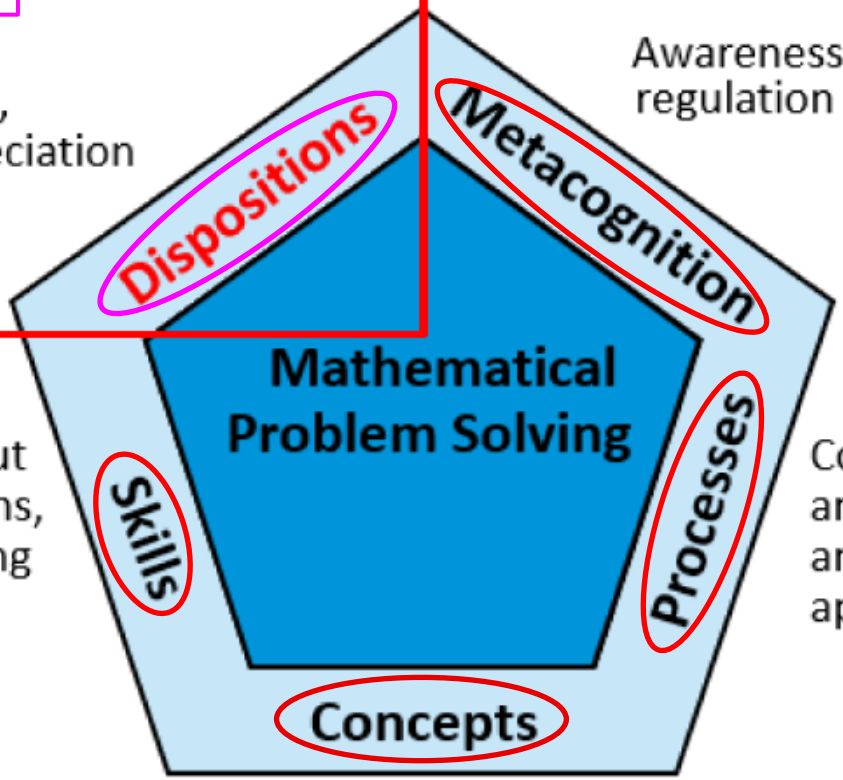
- There must be _____ in worked examples, practice questions and homework questions

Mathematics Curriculum Framework

2020 model

Belief, appreciation, confidence, motivation, interest and perseverance

Curiosity, creativity, adaptability, confidence, perseverance and appreciation



Awareness, monitoring and regulation of thought processes

Proficiency in carrying out operations and algorithms, visualising space, handling data and using mathematical tools

Competencies in abstracting and reasoning, representing and communicating, applying and modelling

Understanding of the properties and relationships, operations and algorithms

Pentagon Model: Attitudes

- _____ is one of the 4 factors that affect math problem solving (Schoenfeld[2], 1985)
- Many students (and even some teachers) believe most (if not all) maths problems can be solved within 5 min
- So what happens if students cannot solve within 5 min? Many of them will just give up!

[2] Schoenfeld, A. H. (1985). *Mathematical problem solving.* Orlando, FL: Academic Press.

Beliefs, Confidence, Perseverance

- Do you really expect to solve most real-life problems in 5 min? Or did you solve most of your university maths problems in 5 min?
- Most genuine maths problems cannot be solved so fast; mathematicians have to _____ for days (or even months and years) to solve them
- This shows how students' beliefs can affect their **confidence** and **perseverance** in solving maths problems

Beliefs, Confidence, Perseverance

- What you can do is to tell and _____ your students that:
  - most genuine maths problems cannot be solved so fast
  - mathematicians have to persevere for days (or even months and years) to solve them
  - share your own university experience of solving maths problems
- Then **encourage and _____ them to persevere**

Appreciation (and Interest)

- How to make students interested and appreciate the beauty and relevance of mathematics?
- Issue 1: Not possible that every student will find same thing interesting (different tastes)
- So **use** _____ **of resources** to interest different students
- Issue 2: Not possible to make every part of every maths lesson interesting
- So your goal is to make **one part of every maths lesson interesting (whenever possible)**

Appreciation (and Interest)

- Providing **enough** _____ for students to be engaged in their hearts: do it often enough
- Issue 3: Pupils may just laugh but learn nothing
- Still have to _____ **it to mathematics**
- **Main purpose** is still not to make students laugh but to make them learn mathematics
- _____

framework[3] to engage hearts of mathematics learners: Linking Opportunities in a Variety of Experiences to the learning of Mathematics

Key Lesson 4

- Need to **engage students'** _____ by making **some** parts of lessons **interesting**

[3] Yeo, J. B. W. (2018). Engaging the hearts of mathematics learners. In P. C. Toh, & B. L. Chua (Eds.), *Mathematics Instruction: Goals, Tasks and Activities* (Association of Mathematics Educators 2018 Yearbook, pp. 115-132). Singapore: World Scientific.

Summary of Learning Theories

- Theory of **constructivism**
- Piaget's assimilation and accommodation
- Skemp's relational understanding
- Bruner's **guided-discovery learning**
- Bruner's C-P-A approach or _____ _____ **approach**
- Theory of behavourism: **"We do one; you do more"** (modified)

Conclusion for this Lecture

- There must be **structure** in worked examples, practice questions and homework questions
- Need to engage students' hearts by making some parts of lessons _____

E-lecture on Use of ICT

- Information on **e-lecture** and **e-quiz** will be available on NTULearn (please don't wait until last minute, in case of technical problems)
- Bruner's C-P-A approach: concrete manipulative → _____ **manipulative** using ICT?

Reflection

- Write down **‘most important’ takeaway** for this lecture that you would like to **remember for a long time** to come